Evidence Grounding and Strategy Evolution for Agent Task Generation from Longitudinal EHRs

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Abstract

Generating high-quality tasks for medical agents from longitudinal EHRs is challenging because patient records differ in the clinical relationships they support, while available tools restrict how evidence can be retrieved. Here, we propose MedEvolve, a framework that improves medical task generation at two complementary levels, task grounding and strategy evolution. At the task-construction level, dual exploration distills recurring clinical-temporal patterns and tool-access constraints into reusable guidance. At the strategy level, feedback-driven evolution then converts verification and independent-solving outcomes into cross-patient construction rules and task-direction preferences, rather than only repairing individual questions. MedEvolve achieves higher verified higher-order task yield than the baselines on both benchmarks. Adding its synthetic tasks improves official-test success rates by 6 and 2 percentage points for ACE and Reflexion, respectively.

1 Introduction

Longitudinal electronic health records (EHRs) document diagnoses, tests, and treatments across encounters and are a primary source of medical agent tasks. MedAgentBench and PhysicianBench evaluate LLM agents in such environments (Jiang et al., 2025; Liu et al., 2026), but task design and review typically rely on clinical experts, limiting scalability. Among automatic generation methods, TaskCraft constructs atomic tasks from document content, verifies them, applies depth and width expansion, and optimizes few-shot prompt configurations using verification pass rates (Shi et al., 2026). CuES and AgentEvolver synthesize tasks from environment interaction trajectories and check feasibility through execution (Mai et al., 2025; Zhai et al., 2025). These approaches construct verifiable tasks from documents or environment interactions. We further examine clinical events, temporal relationships, and evidence patterns in longitudinal records across patients, and use verification and independent-solving feedback to update generation guidance and task-direction weights across patients.

Task generation from longitudinal records faces two challenges. First, task validity depends on both patient records and tool capabilities. A single trajectory does not reveal recurring clinical relationships or evidence distributions across patients, and complete trajectories may exceed the input budget. Whether the required evidence is retrievable must also be checked through tool interaction. Second, different cases support different task types and reasoning demands, so generation guidance needs to incorporate construction issues revealed by evidence verification and independent solving.

We propose MedEvolve, which combines data and environment exploration with feedback-driven strategy updates across patients, without changing model parameters. Our contributions are: (1) grounding task generation in clinical event patterns and tool-access constraints; (2) using evaluation feedback to revise natural-language guidance and task-direction weights; and (3) evaluating valid higher-order task yield in two EHR environments and the downstream value of synthetic tasks for experience learning.

2 Method

2.1 Problem Setting

Given a medical environment $\mathcal{E}$ and a set of longitudinal patient trajectories $\mathcal{D} = \{X_p\}$, the goal is to construct a medical agent task for each trajectory X_p. Each task consists of an instruction q, a candidate reference answer a, and source evidence e. Generation is controlled by a strategy $\theta_r = (g_r, w_r)$, where g_r is natural-language generation guidance, w_r holds sampling weights over task

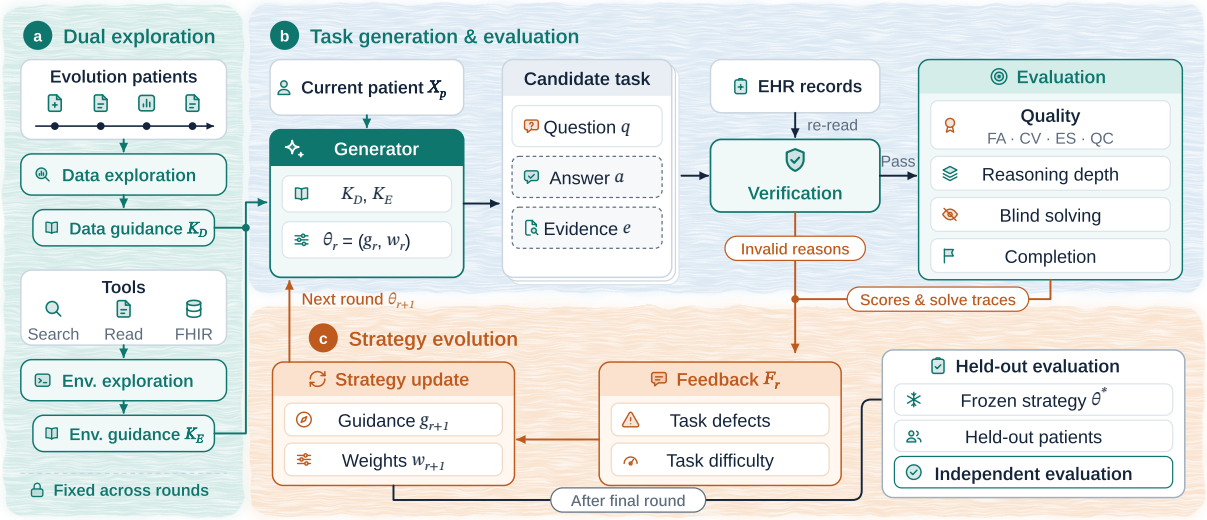


Figure 1: Overview of MedEvolve. (a) Data and environment exploration produce fixed guidance K_D and K_E. (b) Tasks generated from the patient trajectory, exploration guidance, and current strategy undergo source verification, quality and reasoning assessment, independent solving, and completion assessment. Reference answers and evidence claims are withheld from the solver. (c) Evaluation feedback updates generation guidance and direction weights; the final strategy is frozen for held-out evaluation.

types and difficulty directions, and r is the evolution round. MedEvolve first obtains data guidance K_D and environment guidance K_E through exploration, then updates θ_r using evaluation feedback. Figure 1 shows the overall pipeline.

2.2 Data and Environment Exploration

A single patient's trajectory shows only which records that case contains, which makes it hard to tell what pattern those records follow across the whole data source, and reading a large number of patients again at every generation step is infeasible. We therefore run data exploration before generation and organize cross-patient observations into reusable guidance.

Data exploration targets longitudinal patient trajectories. The exploration agent reads the records of multiple patients, analyzes how the data are organized, how they change over time, how clinical events relate to one another, and how evidence is distributed, and distills the information patterns required by different task types into data guidance K_D. K_D describes only reusable patterns and contains no patient-specific clinical facts.

Environment exploration targets the environment interface. The exploration agent actually calls read-only query tools and records access methods, return structures, and interaction constraints into environment guidance K_E, which serves to judge whether the evidence a question requires can be obtained through the tools.

Both explorations are completed during initialization and remain unchanged in all subsequent rounds.

2.3 Task Generation and Evaluation

For patient X_p, the generator conditions on K_D, K_E, and the current strategy θ_r, and generates (q, a, e) according to the specified task type and difficulty direction, where key facts carry references to the original records.

Candidate tasks then pass through verification, scoring, and independent solving in sequence. Verification re-reads the original records and checks that the evidence claims support the reference answer; tasks that pass verification are then assessed for task quality and reasoning depth and answered independently by a fixed agent in the environment. The solving agent has no access to reference answers or evidence claims.

2.4 Feedback-Driven Strategy Updates

After each round, the system aggregates that round's verification defects, quality and reasoning scores, and solve traces, and attributes them into feedback F_r. Attribution must separate defects of the task itself (such as factual errors, insufficient evidence, or ambiguous questions), the genuine reasoning difficulty the task poses, and model, tool, or environment failures, so that execution anomalies are not mistaken for task difficulty. The strategy is

Method	PhysicianBench		MedAgentBench	
	HO count (yield)	Mean level	HO count (yield)	Mean level
MedEvolve	**23 (14.2%)**	2.27	**77 (51.3%)**	**2.73**
EnvExplore	21 (13.0%)	2.25	64 (42.7%)	2.56
TaskCraft	0 (0.0%)	1.01	0 (0.0%)	1.00
Workflow	3 (1.9%)	**2.30**	2 (1.3%)	2.67
SingleShot	3 (1.9%)	1.20	11 (7.3%)	1.44
SelfRefine	4 (2.5%)	2.20	2 (1.3%)	2.00

Table 1: Valid higher-order task generation with GPT-5.4-nano. Higher-order (HO) tasks pass evidence verification and have an aggregate level of at least 3. Yield, shown in parentheses, uses the fixed number of requested tasks as its denominator (162 for PhysicianBench; 150 for MedAgentBench). Mean level covers all verified tasks with difficulty scores. Bold indicates the highest value in each column.

updated as

$$\theta_{r+1} = \mathcal{U}(\theta_r, F_r), \tag{1}$$

where $\mathcal{U}$ is implemented by an LLM. The update acts on two parts: it writes generalizable construction experience into g_{r+1}, adjusting the selection of clinical information, the organization of evidence, and task design; and it adjusts w_{r+1} according to the quality and difficulty performance of each direction, changing the direction distribution of subsequent tasks.

Different rounds use different patient batches to generate tasks. After the scheduled rounds the strategy is frozen and evaluated on a held-out patient set.

3 Experiments

3.1 Evaluation Conditions and Metrics

Environments and comparison conditions. We use 108 PhysicianBench patients and 100 MedAgentBench patients with clinical records, reserving 81 and 75, respectively, for held-out evaluation and the remainder for strategy evolution. The sets are patient-disjoint. All methods share the held-out patients and generation model, GPT-5.4-nano, and generate at most two read-only tasks per patient, yielding fixed totals of 162 and 150 requested tasks. Patient splits and information-access rules appear in Appendix C.1.

Metrics. Tasks are verified against source records. GPT-4o-mini and GPT-5.4-nano independently assess the minimum reasoning required, with scores aggregated into levels 1–4. Verified tasks at level 3 or above count as higher-order (HO). Our primary metrics are HO count and yield, the latter dividing HO count by the fixed requested-task total and thus accounting for generation failures and invalid tasks. Mean level across all verified tasks with difficulty scores is an auxiliary metric. Appendix B details scoring and aggregation.

3.2 Experiment 1: Generating Valid Higher-Order Tasks

We compare generation methods under a shared patient set and task-count limit. SingleShot generates in one pass; Workflow uses localization, generation, and filtering; SelfRefine (Madaan et al., 2023) revises candidates through self-feedback; TaskCraft applies depth and width expansion. EnvExplore uses environment guidance with a fixed initial strategy, whereas MedEvolve combines data and environment exploration with the full strategy after two feedback-update rounds. Computational costs differ across methods; the comparison controls patients and task counts. Appendix C.2 describes procedures and budgets.

MedEvolve achieves the highest HO yield in both environments (Table 1), with a larger margin over the closest baseline, EnvExplore, on MedAgentBench: 13 additional tasks and an 8.6-percentage-point gain, compared with two tasks and 1.2 points on PhysicianBench. The full framework therefore leads in this evaluation, although the PhysicianBench margin is small.

Verification rate and HO yield capture different task properties. TaskCraft produces no HO tasks and has a mean level near 1. MedEvolve has a lower verification rate than some baselines but retains the most valid HO tasks (Appendix D); verification rate alone does not capture reasoning demands. Mean level also requires attention to sam-

ple size: Workflow slightly exceeds MedEvolve on PhysicianBench (2.30 vs. 2.27), based on only 10 verified tasks.

3.3 Experiment 2: Supplementing Official Training Tasks with Synthetic Tasks

We test whether synthetic tasks supplement official tasks for experience learning. In PhysicianBench, ACE (Zhang et al., 2026), Reflexion (Shinn et al., 2023), and OpenClaw (OpenClaw contributors, 2026) each learn under two settings: *Official 20* uses 20 official tasks; *Official + Synthetic 40* adds 20 verified MedEvolve tasks to those same tasks. Both settings use Qwen3.8-27B with fixed model parameters. Experience is frozen after training, and both settings are evaluated on the same 50 official test tasks. Reference answers are used only for scoring. Appendix C.3 details framework mechanisms and reproducibility settings.

Framework	Official 20	Official + Synthetic 40	Δ (pp)
ACE	4.0% (2/50)	10.0% (5/50)	+6.0
Reflexion	12.0% (6/50)	14.0% (7/50)	+2.0
OpenClaw	24.0% (12/50)	20.0% (10/50)	−4.0

Table 2: Task success rates before and after adding synthetic tasks to official training tasks. Success requires the entire task to pass; all rates use the 50 planned test tasks as the denominator. Differences are in percentage points (pp). ACE has 48 scored tasks in the official-only setting and 50 in the mixed setting; all other settings have 50.

Adding synthetic tasks raises success rates by 6.0 percentage points for ACE and 2.0 for Reflexion, but lowers OpenClaw's rate by 4.0 points (Table 2). Synthetic tasks thus show potential in some experience-learning frameworks, with no consistent benefit across the evaluated frameworks. Their different experience representations and update mechanisms may affect task use, but this comparison does not establish the cause of the observed differences.

4 Conclusion

We present MedEvolve, which combines data and environment exploration to ground medical agent task generation in longitudinal patient records and tool-based evidence access. Verification and independent-solving feedback update generation guidance and task-direction weights across patients, without updating model parameters.

On PhysicianBench and MedAgentBench, MedEvolve achieves the highest valid higher-order task yield among the six compared methods, producing 23 and 77 verified higher-order tasks, respectively. Downstream experience-learning experiments further suggest potential value for synthetic tasks in some frameworks: adding synthetic tasks improves official-test success rates for ACE and Reflexion, but reduces the rate for OpenClaw, indicating differences in benefits across frameworks. These findings support dual exploration as an approach to constructing higher-order medical agent tasks and provide an empirical basis for studying how synthetic tasks interact with experience-learning mechanisms.

Limitations

The downstream experience-learning experiment compares training on 20 official tasks with training on those tasks plus 20 synthetic tasks, changing both task count and composition. Controls using an equal number of official tasks or tasks produced by other generation methods would help distinguish the effects of training-set size from those of synthetic-task characteristics. The current downstream results come from a single seed and 50 test tasks in PhysicianBench; the stability of the gains and their performance in other environments require further evaluation.

MedEvolve derives generation guidance from patient records and tool interfaces within a particular environment. The current held-out patient evaluation examines strategy transfer within that environment. The applicability of existing guidance, and the adaptations it requires when healthcare institutions, record structures, or evidence-retrieval tools change, have not yet been systematically evaluated.

References

Yixing Jiang, Kameron C. Black, Gloria Geng, Danny Park, James Zou, Andrew Y. Ng, and Jonathan H. Chen. 2025. MedAgentBench: A virtual EHR environment to benchmark medical LLM agents. *NEJM AI*, 2(9):AIdbp2500144.

Timothy Kassis, Vinayak Agarwal, Yuhuan He, Darshil Patel, and Aubrey M. Brueckner. 2026. Scientific agent skills: A library of procedural knowledge for research agents. *arXiv preprint arXiv:2609.00065*.

Ruoqi Liu, Imran Q. Mohiuddin, Austin J. Schoeffler, Kavita Renduchintala, Ashwin Nayak, Prasantha L. Vemu, Shivam C. Vedak, Kameron C. Black, John L. Havlik, Isaac Ogunmola, Stephen P. Ma, Roopa Dhatt, and Jonathan H. Chen. 2026. PhysicianBench: Evaluating LLM agents in real-world EHR environments. *arXiv preprint arXiv:2605.02240*.

Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegreffe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, Shashank Gupta, Bodhisattwa Prasad Majumder, Katherine Hermann, Sean Welleck, Amir Yazdanbakhsh, and Peter Clark. 2023. Self-Refine: Iterative refinement with self-feedback. In *Advances in Neural Information Processing Systems*, volume 36, pages 46534–46594. Curran Associates, Inc.

Shinji Mai, Yunpeng Zhai, Ziqian Chen, Cheng Chen, Anni Zou, Shuchang Tao, Zhaoyang Liu, and Bolin Ding. 2025. CuES: A curiosity-driven and environment-grounded synthesis framework for agentic RL. *arXiv preprint arXiv:2512.01311*.

OpenClaw contributors. 2026. OpenClaw. Open-source software. Version 2026.9.2; commit 3928bad9badfcb6c7d140530435e806fb8092190.

Dingfeng Shi, Jingyi Cao, Qianben Chen, Weichen Sun, Weizhen Li, Hongxuan Lu, Fangchen Dong, Tianrui Qin, King Zhu, Minghao Liu, Yuchen Eleanor Jiang, Jian Yang, Ge Zhang, Jiaheng Liu, Changwang Zhang, Jun Wang, and Wangchunshu Zhou. 2026. TaskCraft: Automated generation of agentic tasks. In *The Fourteenth International Conference on Learning Representations*.

Noah Shinn, Federico Cassano, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. 2023. Reflexion: language agents with verbal reinforcement learning. In *Advances in Neural Information Processing Systems*, volume 36, pages 8634–8652. Curran Associates, Inc.

Yunpeng Zhai, Shuchang Tao, Cheng Chen, Anni Zou, Ziqian Chen, Qingxu Fu, Shinji Mai, Li Yu, Jiaji Deng, Zouying Cao, Zhaoyang Liu, Bolin Ding, and Jingren Zhou. 2025. AgentEvolver: Towards efficient self-evolving agent system. *arXiv preprint arXiv:2511.10395*.

Qizheng Zhang, Changran Hu, Shubhangi Upasani, Boyuan Ma, Fenglu Hong, Vamsidhar Kamanuru, Jay Rainton, Chen Wu, Mengmeng Ji, Hanchen Li, Urmish Thakker, James Zou, and Kunle Olukotun. 2026. Agentic context engineering: Evolving contexts for self-improving language models. In *The Fourteenth International Conference on Learning Representations*.

A Core Prompts and a Real Example

Figure 2 presents excerpts of the core prompts for dual exploration, task generation, feedback analysis, and strategy updates. The five generation directions are diagnosis and explanation, testing and risk stratification, treatment planning, medication prescribing, and treatment replanning. Two directions are sampled per patient according to their weights.

Data exploration

You are a medical task data exploration agent.

Compressed summaries are not source evidence. Truncated summaries cover only part of the records; omissions do not establish absence.

Identify data organization, temporal changes, clinical relationships, evidence citation patterns, and supported task opportunities.

Return DataExploration. guidance must be abstract and patient-independent. Do not repeat patient IDs, specific facts, or source IDs, or add facts absent from the cache.

Environment exploration

You are a medical task environment exploration agent.

Use native tools to inspect queries, parameters, outputs, and constraints. Access only allowed patients. Query only; do not write or simulate submissions.

Call tools before delivery; a function list does not verify capability. Distinguish verified, failed, and unverified capabilities.

Do not put patient-specific observations or IDs in shared guidance. Do not treat tool or environment failures as task difficulty, or claim unverified capabilities are available.

Task generation

You are a medical task generator.

Inputs: cached patient trajectory, fixed data and environment guidance, current strategy, and two generation directions. Attempt one task for each specified type and difficulty.

Return one TaskBatch with at most two questions. Insufficient evidence permits fewer tasks or an empty list. Do not fabricate. No query tools are available; use only facts and source IDs in the trajectory.

Select facts from original records in trajectory.sources before constructing the question, answer, and evidence.

Each evidence statement must contain only facts supported by its source_id. Cite each source for cross-record conclusions. Put comparisons in reference_answer; state the source facts separately in evidence.

An omitted fact does not mean it is absent, a test was not performed, or a result is normal. Skip a task slot when evidence is insufficient.

Actual subsequent treatment is not the sole answer to a clinical decision task; accept evidence-supported equivalent alternatives.

Feedback analysis

You analyze feedback on medical task generation strategies.

Distinguish quality issues, clinical difficulty, insufficient evidence, ambiguity, and tool, environment, or model failures.

Failures and exhausted budgets do not imply greater difficulty. Without reliable completion assessment, state the limits; do not infer greater difficulty or solver failure.

Recommend changes to topic selection, evidence organization, reference answers, and generation directions.

Strategy update

Return a new Strategy using only the current Strategy and Feedback.

Update generation_guidance and allocation; keep exploration guidance fixed. Experience must be evidence-supported, patient-independent, and retain its applicability conditions.

A task-specific requirement is not automatically a general rule. Allocation assigns two generation slots without changing their number.

Keep the relevant strategy when feedback is insufficient. Do not create difficulty through convoluted wording, missing data, or tool failures.

Figure 2: Excerpts of MedEvolve's core prompts for data exploration, environment exploration, task generation, feedback analysis, and strategy updates.

Figure 3 presents a concise example from an evolution run in PhysicianBench, linking task generation, source-record verification, end-of-round feedback, and strategy updates. Feedback analysis turns verification issues into generation constraints applicable to other patients, rather than merely correcting the current answer. The figure retains key excerpts of the question, candidate answer, and revised guidance to illustrate this process.

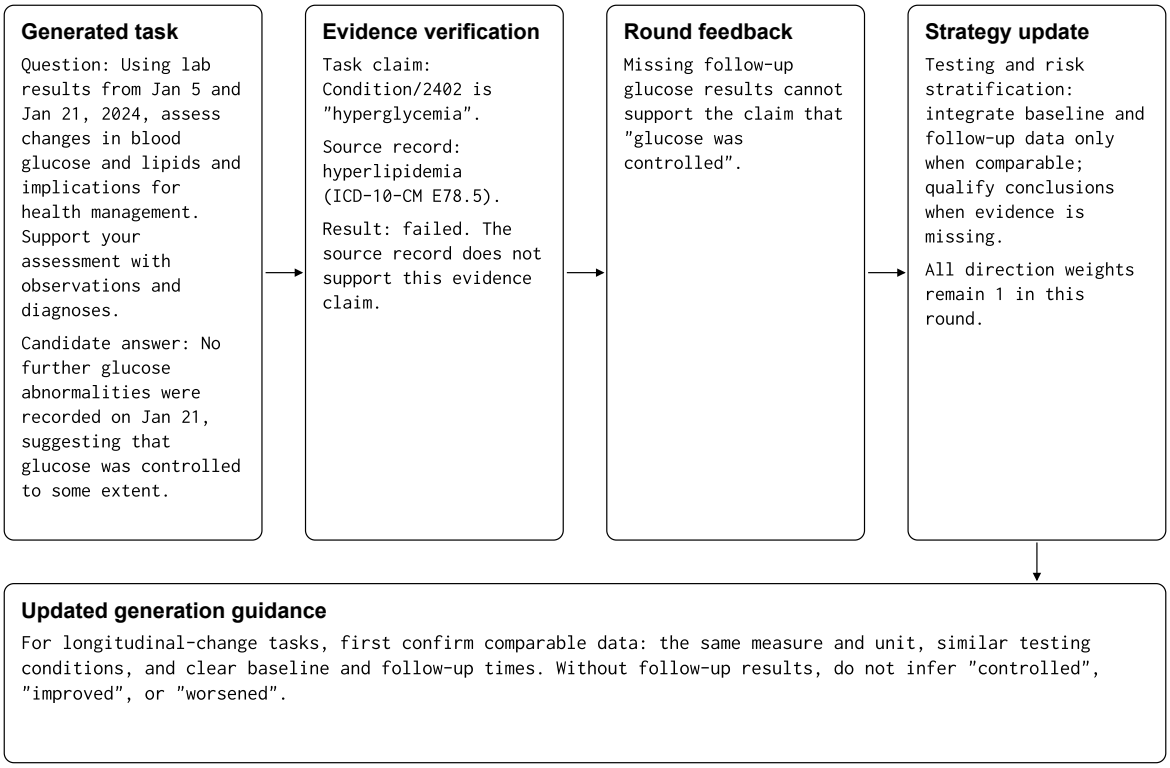


Figure 3: A real example of evidence verification and guidance revision.

B Evidence Verification and Reasoning Assessment

Figure 4 summarizes the key prompts for evidence verification and reasoning-depth assessment, together with score aggregation rules.

Higher-order yield uses the fixed requested-task total as its denominator (162 for PhysicianBench; 150 for MedAgentBench). Generation shortfalls and invalid tasks are not removed from the denominator. Mean level includes only verified tasks with complete difficulty scores.

Evidence-claim verification

Judge only whether this evidence claim is supported by this source record. resource is the original record retrieved using evidence.source_id.

Check each drug name, dose, test value, time, and other fact in statement. Set valid=true only if the record supports the entire claim; set valid=false if contradicted or undocumented.

Do not substitute other records or clinical knowledge for this source. In reason, name the source_id and the supporting or conflicting facts.

Whole-task verification

Verify that the question and reference answer are supported by the original data in evidence_records. Records have been retrieved from the environment, and evidence claims checked individually.

Do not assume patient facts absent from the records. Post-decision data may support the reference answer, but actual subsequent treatment is not the only correct option for a clinical decision task.

Return valid=false for an ambiguous question, insufficient evidence, or an incorrect reference answer, and explain the basis.

Reasoning-depth assessment

Inspect the question, evidence, and candidate answer. Score the simplest sufficient path that correctly meets the core requirements. A candidate answer is neither automatically correct nor the only solution path.

1: Fact retrieval. Read, match, count, summarize, or restate recorded diagnoses. Repeated records do not raise the level.

2: Simple judgment. Facts plus one clinical rule suffice. Multiple independent simple judgments do not imply level 3.

3: Necessary integration. The conclusion depends on at least two independent, indispensable facts and their relationship. Removing a key fact changes or invalidates the conclusion.

4: Decision trade-off. At least two reasonable options, conflicting benefits, risks, or constraints, and patient evidence that determines the choice.

Check shortcuts: does the question state the conclusion, a diagnosis label contain the answer, one record suffice, or generic advice suffice? More evidence, longer time spans, or longer answers alone add no points.

In reason, state the solution path, necessary facts and relationships, shortcuts, and level justification. For level 4, specify options and trade-offs. Do not fabricate unconfirmed patient facts.

Score aggregation and HO tasks

GPT-4o-mini and GPT-5.4-nano score independently. Take the median of two valid scores and round down. If either is missing, no aggregate level is formed.

Verified tasks with aggregate level >=3 are higher-order (HO) tasks.

Figure 4: Prompts for evidence verification and reasoning-depth assessment. Left: verify each evidence claim, then the whole task. Right: reasoning levels 1–4, shortcut checks, and aggregation. The median of two judges' scores is rounded down; verified tasks with an aggregate level of at least 3 count as higher-order tasks.

C Experimental and Reproducibility Settings

C.1 Patient Splits and Information Access

Table 3: Clinical patient pools and splits for two evolution rounds.

Environment	Total	Evol.	Eval.	Round sizes
PhysicianBench	108	27	81	13 / 14
MedAgentBench	100	25	75	12 / 13

Only patients with clinical records enter the pool; MedAgentBench patients with demographics alone are excluded from generation. Patient lists are frozen with seed 42, and patients previously exposed during exploration remain in the evolution set. Evolution and held-out evaluation patients are disjoint. Model inputs are trimmed to the budget, whereas verification reads the original sources. Generation and solving are read-only; time constraints in a question do not imply that tools hide later records.

C.2 Models and Baseline Adaptation

In Experiment 1, GPT-5.4-nano is used for generation, independent solving, verification, and completion assessment. GPT-4o-mini and GPT-5.4-nano judge difficulty. All six methods share the held-out patients, a limit of two tasks per patient, and the common evaluation protocol.

TaskCraft adapts patient records and read-only FHIR tools to conclusion extraction, atomic task generation, verification, and depth/width expan-

sion. The two expansion branches start independently and share the generation budget. Workflow performs localization, generation, and filtering; SingleShot generates in one pass. SelfRefine uses at most three self-reviews and two revisions, stopping early upon acceptance. EnvExplore uses environment guidance and a fixed initial strategy. MedEvolve uses dual exploration and the full strategy after two update rounds. Internal call counts differ, so the comparison does not control for equal computational expenditure.

C.3 Downstream Experience Learning and Official Evaluation

The execution model is Qwen3.8-27B with UD-Q4_K_M quantization, thinking disabled, a budget of 100 model calls per task, and seed 42. Both settings start from the same initial state and share 20 official training tasks. The mixed setting adds verified MedEvolve tasks: 10 generated by GPT-4o-mini and 10 by GPT-5.4-nano. Experience is frozen after training, and both settings are scored on the same 50 official test tasks using PhysicianBench's native checkpoints. Reference answers are used only for scoring.

ACE updates a strategy playbook; Reflexion stores reflections and transferable experience; OpenClaw updates skills through its native Skill Workshop. Reflexion retrieves at most three relevant experiences per task and stores no patient identifiers, individual clinical values, or task-specific answers. Experience is not updated during testing, and scores do not trigger reflection or retries.

Test success rates use all 50 planned tasks as the denominator. ACE completes scoring for 48 tasks in the official-only setting and 50 in the mixed setting; both settings for Reflexion and OpenClaw cover all 50 tasks.

D Supplementary Task Counts

Table 4: Task counts by method in Experiment 1 (generation model: GPT-5.4-nano). Verified denotes tasks that pass evidence verification.

	PhysicianBench		MedAgentBench	
Method	Generated	Verified	Generated	Verified
MedEvolve	157	70	146	104
EnvExplore	141	93	141	108
TaskCraft	128	125	111	108
Workflow	32	10	11	3
SingleShot	162	122	150	127
SelfRefine	12	10	7	6

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